

# Identification of Conflicts in User Story Requirements Using the Clustering Algorithm

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**Abstract**— In a software development life cycle, a critical phase that must be an important concern is the phase of exploring user requirement. In Agile Software Development, user requirements are written in a user story. User stories are written in Natural Language Processing with the format "As I, I want to, So that. There can be conflicts between user story. Grouping user stories will make it easier and faster for software developers to evaluate potentially conflicting or conflicting user stories. The steps are taken starting from preprocessing for syntax checking. These attributes are well formed, atomic, minimal, and unique. The clustering algorithm chosen Single Linkage, and K means. From 31 sentences of user stories passed syntax 115. User stories pass this syntax are then clustered. The results of the evaluation of these two algorithms using the best silhouette values were obtained for single linkage 6 clusters, and k means 6 clusters.

**Keywords:** User Story, Agile development, Inconsistency, Conflict Free, Single Linkage, K means.

## I. INTRODUCTION

The initial stage of the software development life cycle is the requirements engineering stage. This stage is the key to success in the next stages, namely design and coding. The initial stage of requirements engineering is to explore user needs. User requirements are usually written in natural human language, and the use of this language can cause conflict. One approach to software development that is currently widely used is Agile. In agile, requirement sentences are written in a collection of user stories.

User stories are increasingly gaining a place in the software development process, especially in agile software development. User stories are the most widely used artifact in agile software development [1],[2] that express requirements from the user's point of view. A user story is a semi-structured specification of requirements written in natural language. A user story template may take the following form [3]: as [WHO], I want/want to/need/can/would like [WHAT], so that [WHY]. It contains important elements of requirements: WHO wants it, WHAT is expected from the system, and optionally, and WHY it is important [3], [4]. The rise of agile software development has attracted researchers and practitioners into this research field [1], [5], [6]. User stories, as the most widely used artifact in agile software development, are challenging to explore. The fact that they are written in natural language makes them easily understandable to stakeholders. However, requirements written in natural language have drawbacks, such as ambiguity, **inconsistency**, and incompleteness [7][9]. In Requirement Engineering (RE), the term conflict refers to the requirements' inconsistency or interdependency [8][11]. Different studies show that if

conflicts are not successfully resolved, then it can be one of the major reasons for the failure of the product. Any company needs to resolve the conflicts in the early phases of SDLC, or it may reiterate the work from the beginning [10]. Elhassan et al.[13] propose Thus a requirement conflict detection model using machine learning algorithms to automate the classification process of the requirements based on the accuracy rates. Hence, eliminating the conflicting and ambiguous requirements to build a successful IS project based on precise requirements reflects stakeholder needs. Finally, producing such a mechanism will significantly reduce the risk of IS project failure. Thus, increasing the success rates; this grouping system makes it easy to find user stories that allow conflicts. There has been a grouping of user needs sentences but not specific user stories [1]. This grouping is also based on the perspective of stakeholders. No one has done it for grouping user stories to identify conflicts. The results of this grouping will then be evaluated with expert judgment.

This paper discusses grouping user stories using single linkage and K-means clustering algorithms. This grouping aims to localize, simplify and speed up developers in evaluating and correcting conflicting user stories. This paper is organized in the following order. The related work section discusses related research that has existed in grouping needs sentences. The methodology section describes the process stages carried out in this paper. The results of the research and discussion are discussed in the results and discussion section. The final section discusses the conclusions and future research opportunities.

## II. RELATED WORKS

The user story grouping system is a system that combines user stories based on the level of similarity of the user stories. User stories that have elements of similarity have the opportunity to create conflicts or be inconsistent [9][10].

There are various techniques for grouping, including the Single Linkage Method, K-means, and other clustering methods. The single linkage method is a grouping technique that works based on the principle of the Hierarchical Clustering algorithm [1].

The single linkage algorithm has the following stages.

1. Calculate the distance matrix There are various types of distance, but the distance that is often used is Euclidean.

$$d_{ij} = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

Distance formula of Euclidean

Notes:

$d_{ij}$ : Distance between object  $i$  and  $j$

$x_{ij}$ : The value of object  $i$  in the  $-k$  variable

$x_{jk}$ : The value of object  $j$  in the  $-k$  variable

$p$ : The number of observed variables

2. Merge the two closest clusters  
If the distance between objects  $a$  and  $b$  has the smallest distance value compared to the distance between other objects in the Euclidean distance matrix, then the combination of the two clusters in the first stage is  $d_{ab}$ .
3. Update the distance matrix according to the agglomerative clustering method

$$d_{(ab)c} = \min \{d_{a,c}; d_{b,c}\}$$

4. If  $d_{ab}$  is the shortest distance of the Euclidean distance matrix, then the formula for the agglomerative method is:
5. Create Dendrogram

While K-means is a grouping technique based on Partitioned Clustering, the working principle of Hierarchical Clustering is done in stages [1]. Moreover, in each iteration of Hierarchical Clustering, there is only one selection of merging a user story against another user story.

While the working principle of the Partitioned Clustering grouping is to group user stories randomly because the centroid influences them, furthermore, each iteration of the Partitioned Clustering grouping allows for more than one selection of user stories to be combined. [10][12]

K means algorithm:

1. Determine the centroid point as many as K randomly
2. Mark all data based on the nearest centroid point
3. Determine the new centroid point by calculating the average distance of the data in the cluster that has been formed
4. If the centroid point shifts, then the data will be remarked to the new centroid point based on the closest distance
5. Steps 3 and 4 are repeated until the centroid point is no longer shifted

### III. METHODOLOGY

User stories that are grouped in this grouping system are user stories written in English that already use the standard format for writing user stories. The clustering system is carried out using single linkage clustering and k-means clustering methods.

Fig. 1 describes the stages of the research methodology. This methodology starts by entering input as a user story dataset in English, for example: "As a faculty member, I want to access a collection within the repository so that I can obtain research materials." The next stage is pre-processing.

Pre-Processing is a process of checking the User Story in syntax. Each sentence in the user story must meet four syntactical quality attributes: well-formed, atomic, minimal, and unique.

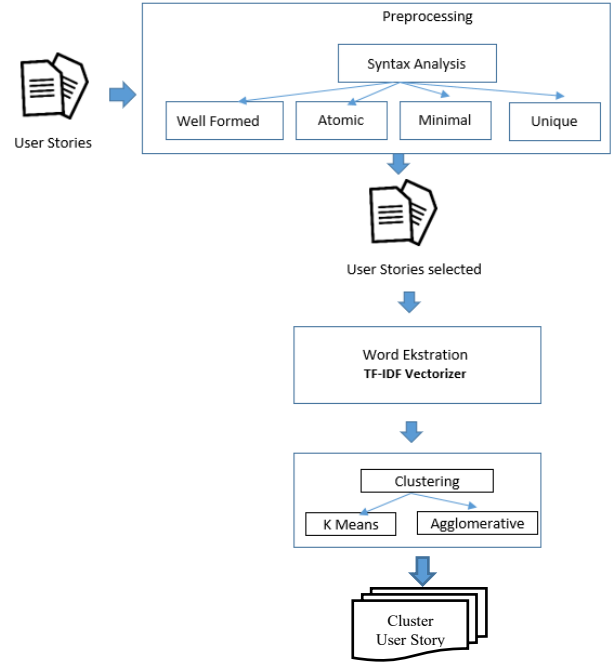


Fig 1. Methodological Step Flow

A user story is considered well-formed if it contains a user story sentence structure; namely there is WHO (As I....) and WHAT (I want/want to....). If one or even both of them are not present, it violates Well Formed.

A user story must be atomic, said to be atomic if and only if there is only one feature, then there can be no symbols or words that violate atomically, such as and, or, /, \, >, <)

A user story must be minimal, meaning it cannot contain elements other than WHO, WHAT, and WHY (optional). Examples that violate the minimum "As a faculty member, I want to direct students to an item within the repository, so that I can facilitate my **teaching (i.e., desires ability to link to content).**"

A user story must be unique, meaning there cannot be a user story like another user story in the SAME project. The output of the pre-processing stage is a user story that has fulfilled the correct syntax, namely well-formed, atomic, minimal, and unique.

The next process is extracting each word to the selected user story in the WHO, WHAT, and WHY sections. First, the data is imported and then cleaned using stopwords, then the data is extracted features by assigning a value to each word based on the frequency of occurrence using TF-IDF Vectorizer TF-IDF is a combination of TF and IDF. TF (Term Frequency) states the number of words in a document, which is logarithmic. The formula used refers to Equation 1 and Equation 2.

$$w_{t,d} = \begin{cases} 1 + \log_{10} tf_{t,d}, & \text{if } tf_{t,d} > 0 \\ 0, & \text{if } tf_{t,d} = 0 \end{cases} \quad (1)$$

While IDF (Inverse Document Frequency) serves to reduce the weight of a word from the number of occurrences in the document.

$$idf_t = \log_{10}(N/df_t) \quad (2)$$

The next process is clustering using the single linkage and K-means methods.

K means Algorithm pseudocode

K-MEANS( $P, k$ )

Input: a dataset of points  $P = \{p_1, \dots, p_n\}$ , a number of clusters  $k$   
Output: centers  $\{c_1, \dots, c_k\}$  implicitly dividing  $P$  into  $k$  clusters

```

1  choose  $k$  initial centers  $C = \{c_1, \dots, c_k\}$ 
2  while stopping criterion has not been met
3    do ▷ assignment step:
4      for  $i = 1, \dots, N$ 
5        do find closest center  $c_k \in C$  to instance  $p_i$ 
6          assign instance  $p_i$  to set  $C_k$ 
7      ▷ update step:
8      for  $i = 1, \dots, k$ 
9        do set  $c_i$  to be the center of mass of all points in  $C_i$ 

```

Clustering using k means refers to the pseudocode.

The Single Linkage algorithm is as follows

Pseudocode of agglomerative clustering for single Linkage

1. Begin with  $n$  clusters, each containing one object, and we will number the clusters 1 through  $n$ .
2. Compute the between-cluster distance  $D(r, s)$  as the between-object distance of the two objects in  $r$  and  $s$  respectively,  $r, s = 1, 2, \dots, n$ . Let the square matrix  $D = (D(r, s))$ . If the objects are represented by quantitative vectors we can use Euclidean distance.
3. Next, find the most similar pair of clusters  $r$  and  $s$ , such that the distance,  $D(r, s)$ , is minimum among all the pairwise distances.
4. Merge  $r$  and  $s$  to a new cluster  $t$  and compute the between-cluster distance  $D(t, k)$  for any existing cluster  $k \neq r, s$ . Once the distances are obtained, delete the rows and columns corresponding to the old cluster  $r$  and  $s$  in the  $D$  matrix, because  $r$  and  $s$  do not exist anymore. Then add a new row and column in  $D$  corresponding to cluster  $t$ .
5. Repeat Step 3 a total of  $n - 1$  times until there is only one cluster left.

#### IV. RESULTS AND DISCUSSION

The requirement sentence used in this study as test data is a requirement sentence using agile methodology. In agile terminology, it is called a user story. A user story has a semi-structured structure which consists of the following words: "As I..... I want/ want t..... So that I..... The data set used in the test was taken from the g27-culrepo.txt dataset with a

total of 119 user stories. G27-culrepo.txt is a data set for the campus administration Information System. One hundred nineteen user stories as a dataset will be eliminated for user stories that violate the quality attributes, namely Well formed, Atomic, Minimal and Unique. Thirty-one user stories pass pre-processing. For example, a user story that passes the syntax quality attribute on the four attributes is "As a faculty member, I want to access a collection within the repository so that I can obtain research materials." Eighty-four user stories did not pass the syntax check. TABLE I is an example of 3 user stories that did not pass the syntax, along with the violated attributes and the reasons. Atomic and Unique, no violations were found in the dataset used.

TABLE I. EXAMPLE OF A USER STORY THAT DOESN'T PASS THE SYNTAX

| User Story                                                                                                                                                                                                                                            | violated attribute | Reasons                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------|
| As a faculty member, I want to direct students to an item within the repository so that I can facilitate my <b>teaching (i.e., desires ability to link to content).</b>                                                                               | Minimal            | There is another element except WHO, WHAT, and WHY (i.e., desires ability to link to content). |
| As a DB/IR administrator, I want to mark report/dataset not to be emailed/viewable due to the error being discovered.                                                                                                                                 | Atomic             | There is notation/, on DB/IR in the part of Who and emailed/viewable under the WHAT            |
| As a cross dept proj team member, I want to have a better place to collect and share project materials, with a range of organization/tagging <b>and</b> access options, including private ( <b>internal group use only</b> ), University, and public. | Atomic, Minimal    | There is "and" and brackets in the parts of WHAT                                               |

In word extraction, namely the word vectorization process with TF IDF in equations 1 and 2, the top 5 word sequences that appear the most in the 31 user stories are "materials, obtain, research, within, and access", Each of these words will be checked for similarities and similarities with all other user stories, namely the first User story (1) to the last User story (31).

Analysis of the results using Silhouette Coefficient value analysis. The silhouette coefficient is obtained by calculating the average distance between user stories in one cluster. After that, the distance between a user story and another user story in another cluster is calculated, and what is taken is the closest distance.

To determine the value of  $k$ , try starting from  $k$  equal to 2, 3, 4 and so on. After evaluating the results,  $k$  with a value of more than 10 will give greater results, but the members in the group are getting smaller, and even some in one cluster contain only 1. The value of  $k$  that will be evaluated is the value of  $k = 2$  to  $k = 10$ .

The clustering process is carried out using a single linkage algorithm with a value of  $k = 2$  to a value of  $k = 10$ . The silhouette value is shown in TABLE II.

TABLE II. SILHOUTTE VALUE RESULT ON SINGLE LINKAGE

| K value  | Silhouette value           |
|----------|----------------------------|
| 2        | 0.04654205685777102        |
| 3        | 0.0633192657988811         |
| 4        | 0.07596371290443668        |
| 5        | 0.09113376132827494        |
| <b>6</b> | <b>0.10116385568227819</b> |
| 7        | 0.09354453123968708        |
| 8        | 0.08955650918802817        |
| 9        | 0.09771909825987574        |
| 10       | 0.10015364331011546        |

Analysis of the silhouette value in TABLE II, for the value of  $k$  from 2 there is an increase in the silhouette value until  $k = 6$  has the highest value of 0.10116385568227819. If the value of  $k$  is added by 1 and so on, there will be an anomaly in the result of the silhouette value, whose value will continue to be high.

The clustering process is carried out using the k-means algorithm with a value of  $k = 2$  to a value of  $k = 10$ . The silhouette value is shown in TABLE III.

TABLE III. SILHOUTTE VALUE RESULT ON K-MEANS

| K value  | Silhouette value            |
|----------|-----------------------------|
| 2        | 0.02158743865058507         |
| 3        | 0.02071634691232585         |
| 4        | 0.04731969955072518         |
| 5        | 0.026706842375457274        |
| <b>6</b> | <b>0.048187124669375886</b> |
| 7        | 0.046581125686211415        |
| 8        | 0.05851321380933133         |
| 9        | 0.056546583846806525        |
| 10       | 0.060123846708168685        |

TABLE III explains the silhouette value for this  $k$  means using the local optimal. that is, by looking at the top of the first graph before it drops as the optimal value. This is because the more clusters there are, the higher the silhouette value. At the first peak,  $k=4$ , the silhouette value is 0.04731969955072518, but when viewed at the second peak,  $k$  equals 6. There is an increase in the silhouette value, so the optimum  $k$  taken is 6.

User story data in the same group for the single Linkage clustering algorithm is shown in TABLE IV.

TABLE IV if you do the analysis manually, as in cluster 6, user stories 21 and 22, with the sentence.

TABLE IV. CLUSTER USER STORIES IN SINGLE LINKAGE

| Cluster | User Story                 |
|---------|----------------------------|
| 1       | 4 5 7 8 27                 |
| 2       | 17 26 31                   |
| 3       | 23 24 30                   |
| 4       | 1 2 3 6 9 14 16 19 20 25   |
| 5       | 10 11 12 13 14 15 18 28 29 |
| 6       | 21 22                      |

US21: As a repository support team member, I want to **move a collection** to fulfill the collection owner's request.

US22: As a repository support team member, **I want to rename a collection** to fulfill the collection owner's request.

In user stories 21 and 22, there is a potential conflict in the WHAT section: move a collection and rename the collection. If you move, the name is the same, it only changes the location, while if you rename it, it will change the name, but the location remains the same.

User story data in the same group for the  $k$  means clustering algorithm is shown in TABLE V.

TABLE V. CLUSTER USER STORIES IN K-MEANS

| Cluster | User Story             |
|---------|------------------------|
| 1       | 4 12 15 18 25 28       |
| 2       | 8 23 24 27 30          |
| 3       | 2 6 9                  |
| 4       | 1 3 5 7 10 16 19 20 31 |
| 5       | 11 13 14 17 26 29      |
| 6       | 21 22                  |

TABLE V for user stories 21 and 22 are also grouped in the same cluster.

## V CONCLUSION

Based on the experiments' results, the identification of potentially conflicting user stories can be grouped into the same cluster. Both clustering algorithms, namely single linkage and  $k$ -means, can group conflicting user stories into the same group. This grouping can appear in the WHO WHAT and WHY sections and their combinations. The manual or expert analysis illustrates that the results of clusters in these two algorithms have similarities with those of experts, for example, in User stories 21 and 22. Further research that can be done is grouping user stories with semantic similarities.

## REFERENCES

- [1] Ishaya Gambo "Identifying and Resolving Conflicts in Requirements by Stakeholders: A Clustering Approach" ENASE 2021
- [2] Marina Sh Murtazina, T.Avdeenko "The detection of conflicts in the requirements specification based on an ontological model and a production rule system" ITNT 2019
- [3] E. M. Schön, J. Thomaschewski, and M. J. Escalona, "Agile Requirements Engineering: A systematic literature review," Comput. Stand. Interfaces, vol. 49, pp. 79–91, 2017, doi: 10.1016/j.csi.2016.08.011.
- [4] R. Noel et al., "Exploring collaborative writing of user stories with multimodal learning analytics: A case study on a software engineering course," IEEE Access, vol. 6, pp. 67783–67798, 2018, doi: 10.1109/ACCESS.2018.2876801.
- [5] Y. Wautelet, S. Heng, M. Kolp, and I. Mirbel, Unifying and extending user story models, vol. 8484 LNCS. 2014.
- [6] G. Lucassen, F. Dalpiaz, J. M. E. M. van der Werf, and S. Brinkkemper, "Improving agile requirements: the Quality User Story framework and tool," Requir. Eng., vol. 21, no. 3, pp. 383– 403, 2016, doi: 10.1007/s00766-016-0250-x.
- [7] I. Inayat, S. Salwah, S. Marczak, M. Daneva, and S. Shamshirband, "A systematic literature review on agile requirements engineering practices and challenges," Comput. Human Behav., vol. 51, pp. 915–929, 2015, doi: 10.1016/j.chb.2014.10.046. Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].

- [8] M. Younas et al., "Elicitation of Nonfunctional Requirements in Agile Development using Cloud Computing Environment," *IEEE Access*, pp. 1–1, 2020, doi: 10.1109/access.2020.3014381.
- [9] H. Meth, M. Brhel, and A. Maedche, "The state of the art in automated requirements elicitation," *Inf. Softw. Technol.*, vol. 55, no. 10, pp. 1695–1709, 2013, doi: 10.1016/j.infsof.2013.03.008.
- [10] A. R. Da Silva, "Linguistic patterns and linguistic styles for requirements specification (I): An application case with the rigorous rsl/business-level language," *ACM Int. Conf. Proceeding Ser.*, vol. Part F1320, no. I, pp. 1–27, 2017, doi: 10.1145/3147704.3147728.
- [11] H. Dar, M. I. Lali, H. Ashraf, M. Ramzan, T. Amjad, and B. Shahzad, "A systematic study on software requirements elicitation techniques and its challenges in mobile application development," *IEEE Access*, vol. 6, pp. 63859–63867, 2018, doi: 10.1109/ACCESS.2018.2874981.
- [12] Sudeep Tanwar, *Resolving Conflicts in Requirement Engineering Through Agile Software Development*, ICICC 2018 Vol 1